

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

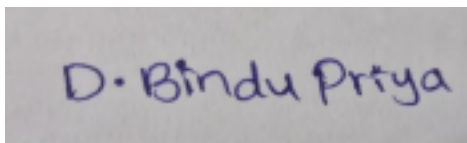
Duration: 90 minutes

Marks: 25

Code of Conduct:

I D.BINDU PRIYA(192425162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Scenario-Based Questions

CO-AT2

QUESTIONS: 05 Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend?

Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured	Mostly organized	Some organization	No logical organization

		hierarchically with logical flow and grouping	with minor structural issues	but lacks clear hierarchy	or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Unsupervised Training of Neural Networks, Auto Encoders

Concept Identification

- Unsupervised Learning
- Autoencoder
- Encoder
- Decoder
- Latent Representation
- Reconstruction Error
- ECG Signal
- Anomaly Detection

Linkages

Unlabeled ECG Signals

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Encoder

↓

Compressed Features

↓

Decoder

↓

Reconstructed ECG

↓

Compare Original and Reconstructed Signal

↓

High Error → Abnormal ECG

Low Error → Normal ECG

Organization

1. Collect unlabeled ECG records.
2. Train an autoencoder using normal ECG patterns.
3. Encode ECG signals into compressed features.
4. Decode the features to reconstruct the ECG.
5. Calculate reconstruction error.
6. Detect abnormalities based on high reconstruction error.

Integration of Literature

- Autoencoders learn useful features without labeled data.
- Representation learning captures hidden ECG characteristics.
- Reconstruction error is commonly used for anomaly detection in healthcare.

Clarity

Recommended Technique: Autoencoder

Justification: Autoencoders automatically learn normal ECG patterns.

Abnormal ECG signals produce higher reconstruction errors, making them

suitable for detecting anomalies without labeled datasets.

2. Auto Encoders, Representation Learning

Concept Identification

- Autoencoder
- Encoder
- Decoder
- Latent Features
- Representation Learning
- Dimensionality Reduction
- Customer Segmentation

Linkages

Customer Purchase Data

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Encoder

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Compressed Features

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Latent Representation

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Clustering Algorithm

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Customer Segments

Organization

1. Collect customer purchase records.
2. Train an autoencoder.
3. Compress the purchase data.

4. Extract latent features.

5. Perform customer segmentation using compressed features.

Integration of Literature

- Autoencoders reduce data dimensions while preserving important information.
- Representation learning improves clustering performance.
- Compressed features remove redundant information.

Clarity

Autoencoders reduce high-dimensional purchase data into compact feature representations. These features improve clustering efficiency and help identify meaningful customer groups.

3. Width and Depth of Neural Networks, Activation Functions

Concept Identification

- Deep Neural Network
- Hidden Layers
- Width
- Depth
- ReLU
- Leaky ReLU
- Softmax
- Feature Learning

Linkages

Face Images



More Hidden Layers

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ReLU Activation

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Hierarchical Feature Extraction

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Softmax Layer

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Face Recognition

Organization

1. Increase the number of hidden layers.
2. Increase neurons where required.
3. Replace sigmoid activation with ReLU or Leaky ReLU.
4. Use Softmax in the output layer.
5. Train the deeper network.

Integration of Literature

- Deep networks learn complex facial features.
- ReLU reduces the vanishing gradient problem.
- Increased depth improves recognition accuracy.

Clarity

A deeper neural network with ReLU activation extracts more meaningful facial features than a shallow network, leading to improved facial recognition performance.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Concept Identification

- Restricted Boltzmann Machine
- Visible Layer
- Hidden Layer
- Collaborative Filtering
- User Preferences
- Recommendation System

Linkages

User-Movie Ratings



Visible Layer



Hidden Features



Preference Learning



Predict Missing Ratings



Movie Recommendations

Organization

1. Create the user-movie rating matrix.
2. Train the RBM using viewing history.
3. Learn hidden user preferences.
4. Predict missing ratings.
5. Recommend movies with the highest predicted ratings.

Integration of Literature

- RBMs perform collaborative filtering without labeled data. •

Hidden neurons capture user interests.

- Learned features improve recommendation quality.

Clarity

RBMs discover hidden relationships between users and movies. They predict unseen preferences and generate personalized movie recommendations without requiring labeled training data.

5. Deep Learning Applications, Representation Learning

Concept Identification

- Representation Learning
- Autoencoder
- Deep Neural Network
- Sensor Data
- Latent Features
- Anomaly Detection
- Defect Detection

Linkages

Sensor Readings



Autoencoder



Latent Features



Reconstruction

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Error Calculation

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Normal Product / Defective Product

Organization

1. Collect sensor readings from products.
2. Train an autoencoder on normal sensor data.
3. Learn compressed feature representations.
4. Reconstruct the sensor readings.
5. Compute reconstruction error.
6. Detect defective products based on high reconstruction error.

Integration of Literature

- Representation learning extracts important sensor patterns.
- Autoencoders detect anomalies using reconstruction loss.
- Deep learning enables real-time industrial quality monitoring.

Clarity

An autoencoder-based representation learning model learns the normal behavior of sensor readings. Products with high reconstruction errors are classified as defective, enabling accurate and efficient quality inspection.